

Hamza Akmal Chaudary

✉ chaudary.h@northeastern.edu | 📠 hamza-akmal-ch | 🌐 hamzaakmal98

EDUCATION

MS Artificial Intelligence — *Northeastern University* SEPT 2024 - MAY 2026
Alfond Scholarship for AI Recipient (\$25,000)
Relevant Coursework: Software Engineering, Databases, LLMs as Dialogue Agents, Programming Design Paradigms, Algorithms

BS Computer Science - Physics — *Lahore University of Management Sciences* SEPT 2020 - MAY 2024
Graduated with High Merit - Dean's Medal for Best Senior Project (Deep Neural Networks in Quantum Mechanics)
Relevant Coursework: Deep Learning, Computer Vision, Quantum Computing, Data Structures, NLP, Quantitative Finance

EXPERIENCE

Massachusetts Institute of Technology (MIT) — *Graduate Research Assistant* OCT 2025 - PRESENT
◦ **Research Work:** Our research involves Quantum Error Correction (QEC) codes capable of handling biased and erasure errors. By leveraging engineered biased Pauli noise models and erasure detection techniques, we aim to increase the QEC threshold and reduce resource overhead through the co-design of QEC codes and gate operations to build fault tolerant quantum computers.

Fidelity Investments — *Quantitative Development Intern (Summer 2025)* JUN 2025 - AUG 2025
◦ **C/C++ Programming:** Deployed 4500+ lines to production on Fidelity's new data service client and shared library testing framework (Aurora). Worked directly alongside the Principle Quant Developer at the Boston HQ to brainstorm, design and optimize Clang 20 compiler bindings to configure network protocols that facilitated 18 million trader API requests daily and slashed overnight batch processing times by 43%. Further built a ubiquitous testing service wrapper in python that exposed 18 C++ shared libraries in memory using pybind11.
◦ **Generative AI Tool Development:** Spearheaded an initiative to train an inhouse LLM (Genie) trained on Fidelity Internal records, client interviews and portfolio manager trades over the past 22 years. Fine-tuned Falcon-40b and developed an end-to-end web app incorporating a code-generation tool (1.6 million lines of code generated over the first month) and an interactive chat interface (queried by over 15,000 employees and generated 13 billion tokens over the first month). Awarded Intern Project of the Summer for the task.

Amazon Web Services (AWS) — *Machine Learning Engineer (Contract)* FEB 2023 - SEPT 2024
◦ **ML Model Deployment:** Engineered and productionized a Geospatial Convolutional Neural Network (GCNN) to detect and localize over 152,000 brick kilns across South Asia and North Africa using multispectral satellite imagery (Sentinel-2, Landsat 8). Integrated inference pipelines with AWS Lambda, S3, and EC2 for scalable distributed processing. Delivered results via a geo-referenced ArcGIS REST API, supporting 3,200+ daily user requests and enabling continuous pollutant emission monitoring through spectral signature analysis.
◦ **Web App Deployment:** Worked with the University of Oxford's AI for Science Lab to build the APAD (Air Pollution Asset Database) platform using the MERN stack (MongoDB, Express, React, Node.js), with serverless backend data ingestion. Designed a custom geospatial query engine leveraging Turf.js and AWS Lambda for dynamic pollutant geolocation. Frontend supported real-time asset detection using Leaflet and D3.js, with satellite data streams versioned and cached on Amazon S3 for low-latency rendering.
◦ **Policy-Focused Geospatial Tooling:** Built interactive choropleth and heatmap layers in ArcGIS and Mapbox GL for data-driven policy impact analysis and donor communication. Developed pollutant concentration scoring algorithms that integrated historical weather and emissions data. Raised \$45,000 in funding from Nature Finance by demonstrating quantifiable ROI on a pilot air purification system for brick kiln regions in Pakistan and Bangladesh. Data for the purifiers is asynchronously loaded and available through our live database.

Afiniti — *Machine Learning Engineer Intern* MAY 2022 - SEPT 2022
◦ **ETL Pipeline Optimization:** Designed and deployed a fault-tolerant ETL architecture integrating SAS macros, SQL stored procedures, and Qualtrics REST APIs to automate ingestion, validation, and normalization of high-volume client survey datasets. Implemented dynamic schema mapping and anomaly detection modules, reducing ETL latency by 40% and contributing to a 10bps ROI uplift through improved downstream resource reallocation.
◦ **Distributed Data Processing:** Provisioned and tuned distributed compute clusters using Apache Spark (PySpark) and Dask across AWS EMR and on-premise Kubernetes environments to process terabyte-scale behavioral telemetry data. Applied custom partitioning strategies and memory-aware DAG optimizations, cutting job runtimes by 24% and generating annual compute savings of \$130K through reduced overprovisioning and idle node utilization.
◦ **Real-Time Video Analytics Pipeline:** Engineered a high-throughput video analytics pipeline leveraging OpenCV for batch frame extraction and TensorFlow Object Detection API (YOLOv5, SSD) for temporal event recognition in UX test footage. Processed 500+ hours of video at 30 FPS with 92% mAP accuracy in detecting click, scroll, and hover events, enabling actionable user journey insights.

MCB — *Quantitative Analyst Intern* MAY 2020 - SEPT 2021
◦ **Quantitative Research:** Developed and backtested algorithmic trading strategies on Pakistan Stock Exchange (PSX) using proprietary high-frequency research infrastructure written in C++. Engineered alpha signals by analyzing tick-level order book and historical price-volume data, incorporating momentum, mean-reversion, and volatility-based features. Conducted parameter optimization and walk-forward testing to evaluate strategy robustness across multiple market regimes.

TECHNICAL SKILLS AND ACHIEVEMENTS

- **Languages:** Python, C/C++, Java, Golang, React, Rust, SQL, Typescript/Javascript, Haskell
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Word2Vec, BeautifulSoup, MERN Stack, Next.js, React.js, Node.js, GraphQL
- **Tools & Platforms:** AWS, Google Cloud, Docker, Git, OAuth, SSO, Neo4j, ChromaDB, PostgreSQL, MongoDB, OpenWPM, ArcGIS
- **Honors & Awards:** Recipient of \$250,000 Cornelius Vanderbilt Merit Scholarship; 7× Hackathon Winner (including MLH and university-level competitions); Winner, NASA International Space Settlement Design Competition; Winner, Microsoft + MIT AI for Healthcare Hackathon 2024; Dean's Gold Medal for Senior Thesis.